

Comprehensive Gröbner Systems

1 Introduction

2 Comprehensive Gröbner systems and bases

Let K and L be two fields, such that L is an algebraic closure of K . Let $\mathbf{X}$ and $\mathbf{U}$ be two independent sets of variables, respectively of cardinality n and m . We will refer to $\mathbf{X}$ as *(main) variables* and to $\mathbf{U}$ as *parameters*. $T(\mathbf{X})$, $T(\mathbf{U})$, and $T(\mathbf{X}, \mathbf{U})$ indicate, respectively, the terms using only the variables $\mathbf{X}$, only the parameters $\mathbf{U}$, and both the variables and the parameters. Fix a term order $<_{\mathbf{X}, \mathbf{U}}$ on $K[\mathbf{U}, \mathbf{X}]$ such that $\mathbf{X} \gg \mathbf{U}$, i.e. any term in $T(\mathbf{X})$ is greater than any term in $T(\mathbf{U})$, and denote $<_{\mathbf{X}}$ its restriction on $K[\mathbf{X}]$.

For a polynomial $f \in K[\mathbf{U}, \mathbf{X}]$, call the *leading term (coefficient, monomial) of f under $\mathbf{U}$* the leading term (coefficient, monomial) of f as a polynomial with coefficients in $K[\mathbf{U}]$. These three objects are denoted, respectively, by $\text{LT}_{\mathbf{U}}(f)$, $\text{LC}_{\mathbf{U}}(f)$, and $\text{LM}_{\mathbf{U}}(f)$.

Let $\mathbf{a} \in L^m$, define the canonical specialization homomorphism $\sigma_{\mathbf{a}} : K[\mathbf{U}] \rightarrow L$ induced by $\mathbf{a}$. It can be naturally extended to $\sigma_{\mathbf{a}} : K[\mathbf{U}][\mathbf{X}] \rightarrow L[\mathbf{X}]$. Let $F = f_1, \dots, f_k \subseteq K[\mathbf{U}]$, its zero set is

$$V(F) = V(f_1, \dots, f_k) = \{\mathbf{a} \in L^m : f_1(\mathbf{a}) = \dots = f_k(\mathbf{a}) = 0\} \subseteq L^m.$$

An *algebraically constructible subset of L^m* , is a subset of L^m that can be realized as a finite combination of zero sets of polynomial families $K[\mathbf{U}]$. For example, given $f_1, \dots, f_k, g_1, \dots, g_l \in K[\mathbf{U}]$, $V(f_1, \dots, f_k) \setminus V(g_1, \dots, g_l)$ is an algebraically constructible subset of L^m .

Definition 1. Let F be a subset of $K[\mathbf{U}, \mathbf{X}]$, $\mathcal{A}_1, \dots, \mathcal{A}_l$ be algebraically constructible subsets of L^m and $G_1, \dots, G_l$ be subsets of $K[\mathbf{U}, \mathbf{X}]$. Let $\mathcal{S}$ be a subset of L^m such that $\mathcal{S} \subseteq \mathcal{A}_1 \cup \dots \cup \mathcal{A}_l$. $\mathcal{S}$ is called the *parameter space*. A finite set $\mathcal{G} = \{(\mathcal{A}_1, G_1), \dots, (\mathcal{A}_l, G_l)\}$ of pairs is called *comprehensive Gröbner system (CGS) on $\mathcal{S}$ for F* if $\sigma_{\mathbf{a}}(G_i)$ is a Gröbner basis of the ideal $\langle \sigma_{\mathbf{a}}(F) \rangle$ in $L[\mathbf{X}]$ for each $i = 1, \dots, l$ and $\mathbf{a} \in \mathcal{A}_i$. Each $(\mathcal{A}_i, G_i)$ is called *segment of $\mathcal{G}$* . We simply say $\mathcal{G}$ is a *comprehensive Gröbner system (CGS) for F if $\mathcal{S} = L^m$* .

If a segment has the form (E, N, G) , with E, N subsets of $K[\mathbf{U}]$, the associated algebraically constructible set is $V(E) \setminus V(N)$. With an abuse of terminology, the word “segment” is used in the following for both the couple $(\mathcal{A}_i, G_i)$ and for the constructible set $\mathcal{A}_i \subseteq L^m$.

Definition 2. Given an ideal $I \subseteq K[\mathbf{U}, \mathbf{X}]$, and $S \subseteq L^m$ the *parameter space*, a finite set $\mathcal{G} \subseteq K[\mathbf{U}, \mathbf{X}]$ is called *comprehensive Gröbner basis (CGB) of I on S* , if for all $\mathbf{a} \in S$, $\sigma_{\mathbf{a}}(\mathcal{G})$ is a Gröbner basis of the ideal $\sigma_{\mathbf{a}}(I)$ on $L[\mathbf{X}]$. We simply say $\mathcal{G}$ is a *CGB of I if $S = L^m$* .

A CGB $\mathcal{G}$ of a parametric ideal I is called *minimal* if no proper subset of $\mathcal{G}$ is also a CGB of I .

Remark 3. Notice that, in contrast to the standard definition of Gröbner basis, neither a CGB or the sets G_i of a CGS are required to be a subset of I .

Definition 4. A CGB of a parametric ideal I is *faithful* if the CGB is a subset of I . In the same way, a CGS for a parametric system F is *faithful* if each set G_i , $i = 1, \dots, l$, is contained in $\langle F \rangle$.

Example 5. Consider the ideal $I = \langle a^2x \rangle$, it can be verified that each set $\{a^i x\}$, for $i > 0$ is a CGB. For each $i \geq 2$, the CGB is a subset of I . However, $\{ax\}$ is not faithful as $ax \notin \langle a^2x \rangle$.

Proposition 6. Given a faithful CGS of a parametric ideal I , the set of all polynomials appearing in various segment of the CGS constitute a CGB of I . The CGB so generated is also faithful.

3 Algorithms

3.1 Suzuki–Sato algorithm for CGBs

The Suzuki–Sato algorithm for computing comprehensive Gröbner bases consists of two parts. The first constructs *faithful* comprehensive Gröbner systems.

Notation (Add a variable). Suppose we have finite sets of polynomials $F \subseteq K[\mathbf{U}, \mathbf{X}]$ and $S \subseteq K[\mathbf{U}]$ of polynomials. We introduce a new auxiliary variable λ , and fix a term order $<_{\lambda, \mathbf{X}, \mathbf{U}}$ on $T(\lambda, \mathbf{X}, \mathbf{U})$ that extends $<_{\mathbf{X}, \mathbf{U}}$ such that $\lambda \gg \mathbf{X} \gg \mathbf{U}$. Let $f \in K[\lambda, \mathbf{U}, \mathbf{X}]$. We write $\text{LT}(f)$, $\text{LC}(f)$, $\text{LM}(f)$ for the leading term, coefficient, monomial with respect to $<_{\lambda, \mathbf{X}, \mathbf{U}}$ respectively. As in the previous section, the notation $\text{LT}_{\mathbf{U}}(f)$, $\text{LC}_{\mathbf{U}}(f)$, $\text{LM}_{\mathbf{U}}(f)$ refers to the leading term, coefficient, monomial of the leading term of f with respect to $<_{\lambda, \mathbf{U}, \mathbf{X}}$ as a polynomial in $(K[\mathbf{U}])[\lambda, \mathbf{X}]$. We define homomorphisms

$$\sigma^i : K[\lambda, \mathbf{U}, \mathbf{X}] \rightarrow K[\mathbf{U}, \mathbf{X}], \quad f(\lambda, u, x) \mapsto f(i, u, x) \quad \text{where } i \in \{0, 1\},$$

that simply evaluate the polynomial at $\lambda = 0$ and $\lambda = 1$ respectively

In general, given a collection of polynomials $F \subseteq R$ in a polynomial ring R , then there is an algorithm $\text{REDUCEDGRÖBNERBASIS}(F, <)$ that returns the reduced Gröbner basis of $\langle F \rangle$ with respect to a term order $<$. Recall that this set is the unique minimal Gröbner basis whose leading coefficients are one.

Algorithm 7 ([2, Algorithm CGBMain]). *Let $F \subseteq K[\mathbf{U}, \mathbf{X}]$ be a finite subset and $S \subseteq K[\mathbf{U}]$ a finite subset such that $V(S) \subseteq V(\langle F \rangle \cap K[\mathbf{U}])$, the following algorithm produces a faithful comprehensive Gröbner system on $V(S)$ for F .*

CGBMAIN(F, S):
if $1 \in \langle S \rangle$ **then**:
 return $\emptyset$;
 $A := \lambda \cdot F \subseteq K[\lambda, \mathbf{U}, \mathbf{X}]$;
 $B := (\lambda - 1) \cdot S \subseteq K[\lambda, \mathbf{U}, \mathbf{X}]$;
 $G := \text{REDUCEDGRÖBNERBASIS}(A \cup B, <_{\lambda, \mathbf{X}, \mathbf{U}})$;
 $\{h_1, \dots, h_\ell\} := \{\text{LC}_{\mathbf{U}}(g) : g \in G, \text{LT}(g) \notin T(\mathbf{X}, \mathbf{U}), \text{LC}_{\mathbf{X}, \mathbf{U}}(g) \notin K[\mathbf{U}]\}$;
 (form of h_i is $g = h_i \lambda t + g'$ where $t \in T(\mathbf{X})$ and $\text{LT}_{\mathbf{U}}(g) = \lambda t$)
 $h := \text{lcm}\{h_1, \dots, h_\ell\}$;
return $(S, \{h\}, \sigma^1[G]) \cup \bigcup_{i=1}^\ell \text{CGBMAIN}(F, S \cup \{h_i\})$;

This gives us the following algorithm for the comprehensive Gröbner basis.

Algorithm 8 ([2, Algorithm CGB]). *Let $F \subseteq K[\mathbf{U}, \mathbf{X}]$ be a finite subset. The following algorithm produces a comprehensive Gröbner basis for F .*

CGB(F):
 $S := \text{GENERATINGSET}(\langle F \rangle \cap K[\mathbf{U}])$; (Elimination)
 $\mathcal{G} := \text{CGBMAIN}(F, S)$;
 $G := S$;
for each $(S', T', G') \in \mathcal{G}$ **do**:
 $G = G \cup G'$;
return G

Example 9. *Let $<$ be the lexicographic term order with $x > y > z$. Compute a comprehensive Gröbner system / basis of*

$$F = \{x^3 - a, y^4 - b, x + y - z\}$$

where a, b are parameters. From this we get the minimal polynomial of z .

The implementation in [2] gives a Gröbner system with seven segments. Below are the segments with non-vanishing and vanishing polynomials in a, b , and the minimal polynomial of z .

1. $((b)(a)(729a^4 - 4096b^3)(729a^4 + 64b^3)(16767a^4 + 5632b^3) \neq 0)$
 $-z^{12} + 4az^9 + 3bz^8 - 6a^2z^6 + 48baz^5 - 3b^2z^4 + 4a^3z^3 + 30ba^2z^2 + 12b^2az - a^4 + b^3$
2. $((a) \neq 0, b = 0)$

- $-z^{12} + 4az^9 - 6a^2z^6 + 4a^3z^3 - a^4$
 3. $(b = 0, a = 0)$
 z^6
 4. $((b) \neq 0, a = 0)$
 $-z^{12} + 3bz^8 - 3b^2z^4 + b^3$
 5. $((b)(a) \neq 0, 729a^4 - 4096b^3 = 0)$
 $-52488a^2z^{11} + 46656ba^2z^{10} - 41472b^2z^9 + 216513a^3z^8 - 34992ba^2z^7 + 31104b^2az^6 - 1797120b^3z^5 +$
 $2803734ba^3z^4 - 2649672b^2a^2z^3 + 3534912b^3az^2 + 5705216b^4z - 272727b^2a^3$
 6. $((b)(a) \neq 0, 729a^4 + 64b^3 = 0)$
 $-6561a^2z^{10} - 2916ba^2z^9 - 1944b^2z^8 + 39366a^3z^7 + 39366ba^2z^6 + 21384b^2az^5 + 16848b^3z^4 +$
 $205578ba^3z^3 + 52731b^2a^2z^2 + 41436b^3az + 6344b^4$
 7. $((b)(a) \neq 0, 16767a^4 + 5632b^3 = 0)$
 $-16767z^{12} + 67068az^9 + 50301bz^8 - 100602a^2z^6 + 804816ba^2z^5 - 50301b^2z^4 + 67068a^3z^3 +$
 $503010ba^2z^2 + 201204b^2az + 22399b^3$

There are a few more families of examples listed in [2]. For instance one can vary the powers of x and y in the above system, e.g., $F = \{x^4 - a, y^5 - b, x + y - z\}$. However, for these other examples the expected results are not written out; only the timings.

Example 10. *Ollie: I'm quite interested in the following family of examples coming from Rigidity Theory. Let $G = (V, E)$ be a graph and fix a positive integer d . Consider the parametrised polynomial systems $F = \{f_e\}_{e \in E} \subseteq K[\lambda_e : e \in E][x_{v,k} : v \in V, 1 \leq k \leq d]$, where $K \in \{\mathbb{C}, \mathbb{R}\}$ and*

$$f_{ij} = \sum_{k=1}^d (x_{i,k} - x_{j,k})^2 - \lambda_{ij}.$$

Explicitly, the variables are $x_{v,k}$ and the parameters are λ_{ij} . I would like to know what is the comprehensive Gröbner system for F on the torus $\{\lambda_e \neq 0\}$. I'm mostly intereted in what the regions of the CGB look like and how they are related to so-called 'flexible realisations' of G .

Concrete example 1. *Suppose G is K_4 minus an edge, which has vertex set $\{1, 2, 3, 4\}$ and edges:*

$$E(G) = \{12, 13, 23, 14, 24\}.$$

The system F is generically zero dimensional and, while I do not know for certain, I'm hoping to see very few families in the CGB; and essentially nothing if we insist that $\lambda \neq 0$.

Concrete example 2. *Suppose G is the prism graph on vertex set $\{1, \dots, 6\}$ and edges:*

$$E(G) = \{12, 13, 23, 45, 46, 56, 14, 25, 36\}.$$

The system F is generically zero dimensional. I would like to know what Gröbner bases look like away from this generic region where none of the λ_e is zero.

3.2 Kapur algorithm for CGBs

The algorithm proposed by Kapur in [1] is a completion algorithm. It tries to parallel the Buchberger algorithm in the parametric case.

Algorithm 11 ([1, Algorithm Completion]). *Let F a finite set in $K[\mathbf{U}, \mathbf{X}]$ and (E, N) the parameter space, where $E, N \subseteq K[\mathbf{U}]$. The following algorithm produces a minimal comprehensive Gröbner basis $\mathcal{G}$ of the ideal $I = \langle F \rangle$.*

COMPLETION(F, E, N):

$\mathcal{G} := F;$

if $|\mathcal{G}| \leq 1$ **then return** $\mathcal{G}$: **endif**

for each $f \in \mathcal{G}$:

if REDUNDANT($f, \mathcal{G}, E, N$) **then**

$\mathcal{G} := \mathcal{G} \setminus \{f\};$

else $\Phi_f := \text{SPECIALIZEDTUPLE}(f, E, N);$

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    endif
endfor
 $P := \{\pi = (f_\alpha, g_\beta, \alpha \cap \beta) \mid \alpha \cap \beta \neq \emptyset, f, g \in \mathcal{G}, (f_\alpha, \alpha) \in \Phi_f, (g_\beta, \beta) \in \Phi_g, \text{NECESSARY}(\pi)\};$ 
while  $P \neq \emptyset$  do
    Choose  $\pi = (f_\alpha, g_\beta, \delta_{ij}) \in P$ ;
     $h_{ij} = \text{S-POLYP}_{\delta_{ij}}(f, g)$ ;
     $h_{ij} \xrightarrow[\mathcal{G}]{\delta_{ij}^*} r$ ;
    if  $r$  is determined to be 0 under  $\delta_{ij}$  then
         $P := P \setminus \{\pi\}$ ;
    else
        for each  $q \in \mathcal{G}$ :
            if  $\text{REDUNDAND}(q, \mathcal{G} \cup \{r\}, E, N)$  then
                 $\mathcal{G} := \mathcal{G} \setminus \{q\}$ ;
                 $P = P \setminus \{(\phi_\gamma, \psi_\eta, \gamma \cap \eta) \mid \phi = q \text{ or } \psi = q\}$ ;
            endif
        endfor
         $\Phi_r := \text{SPECIALIZEDTUPLE}(r, E, N)$ ;
         $P := P \cup \{(r_\gamma, \phi_\eta, \gamma \cap \eta) \mid \gamma \cap \eta \neq \emptyset, \phi \in \mathcal{G}, (r_\gamma, \gamma) \in \Phi_r, (\phi_\eta, \eta) \in \Phi_\phi\}$ ;
         $\mathcal{G} := \mathcal{G} \cup \{r\}$ ;
    endif
endwhile
return  $\mathcal{G}$ ;

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3.2.1 Tuple representation

Definition 12. Given a segment $\alpha \subseteq L^m$, a parametric polynomial $f \in K[\mathbf{U}, \mathbf{X}]$ with respect to α is the couple $f_\alpha = ([\bar{f}, \underline{f}], \alpha)$, where $\bar{f}$ and $\underline{f}$ are respectively the non-zero and zero parts of f while evaluating in α .

Notice that $f = \bar{f} + \underline{f}$. Denote $\text{LT}(f, \alpha)$ and $\text{LC}(f, \alpha)$ the leading term and the leading coefficient of a parametric polynomial f under a segment α .

Definition 13. A polynomial $f \in K[\mathbf{U}, \mathbf{X}]$ is called ambiguous under α if its leading coefficient is not determined under α , i.e. $\text{LC}(f, \alpha) \notin L[\mathbf{X}]$. Otherwise, f is called unambiguous.

Example 14. Let $\mathbf{U} = \{a, b\}$, $\mathbf{X} = \{x, y\}$, and $f = ab^2y^2 + b^3 - 1$ and $\alpha = (\emptyset, \{a^3b^2 + ab^3 - a\})$. It is easy to see that f is ambiguous under α . For solve the ambiguity, one has to partition α :

$$i) \alpha_1 = \alpha \cup \{ab^2 \neq 0\}$$

$$ii) \alpha_2 = \alpha \cup \{ab^2 = 0\}.$$

Definition 15. Given a polynomial $f = p_1(\mathbf{U})m_1(\mathbf{X}) + \dots + g_\ell(\mathbf{U})m_\ell(\mathbf{X})$, where $m_1 > \dots > m_\ell$, let α_k be the segment $(\{p_{k-1}\}, \{p_k\})$, for $k = 1, \dots, \ell$, where $\{p_{-1}\} = \emptyset$. The tuple Φ_f is the finite set of tuples $\{f_{\alpha_1}, \dots, f_{\alpha_n}\}$

Notice that the segments $\alpha_1, \dots, \alpha_\ell$ defined as in the previous definition are disjoint and form a partition of the parameter space.

Example 16. As in the previous example let $f = ab^2y^2 + b^3 - 1$. Then

$$\begin{aligned} \Phi_f = \{ & (f_1 = [ab^2y^2 + b^3 - 1, 0], \alpha_1 = (\emptyset, \{ab^2\})) \\ & (f_2 = [b^3 - 1, ab^2y^2], \alpha_2 = (\{ab^2\}, \{b^3 - 1\})) \}. \end{aligned}$$

The tuple $([0, f], (\{ab^2y^2, b^3 - 1\}, \{1\}))$ corresponding to the specialized polynomial 0 is omitted.

Algorithm 17. Let f a parametric polynomial in $K[\mathbf{U}, \mathbf{X}]$ and (E, N) the parameter space, where $E, N \subseteq K[\mathbf{U}]$. The following algorithm return the tuple Φ_f defined in (15).

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SPECIALIZEDTUPLE( $f, E, N$ ):
    return  $\Phi_f$ 

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3.2.2 Parametric rewriting

Parametrically rewriting, or reducing, a parametric polynomial with respect to a parametric basis under a segment has some desired proprieties. A polynomial, even if in the ideal, should not parametrically reduce to 0 using a basis if there is some specialization for which it does not reduce to 0 using the associated specialized basis ($B = \{f = uy + x, g = vz + x + 1\}, h = vz - uy + 1$, specialize to $\{u = 0, v = 0\}$; classically reduces to 0, parametrically it should be a normal form wrt B).

Definition 18. *Given two distinct parametric polynomials $f, g \in K[\mathbf{U}, \mathbf{X}]$, let $f_\alpha = ([\bar{f}, f], \alpha)$ and $g_\beta = ([\bar{g}, g], \beta)$ be the unambiguous forms of f under the segment α and g under the segment β respectively. The a parametric S -polynomial of f and g under a non-empty segment $\alpha \cap \beta$ is*

$$\text{S-POLYP}_{\alpha \cap \beta}(f_\alpha, g_\beta) = \frac{\text{lcm}(\text{LM}(\bar{f}), \text{LM}(\bar{g}))}{\text{LM}(\bar{f})} f - \frac{\text{lcm}(\text{LM}(\bar{f}), \text{LM}(\bar{g}))}{\text{LM}(\bar{g})} g.$$

Remark 19. *Following the notation of the previous definition, let the tuple $\pi = (f_\alpha, g_\beta, \alpha \cap \beta)$. In the Algorithm 11, the condition $\text{NECESSARY}(\pi)$ checks that the following conditions are not met: can be discarded, i.e. the related S -polynomial reduces trivially to 0 under $\alpha \cup \beta$, if:*

- $\text{GCD}(\text{LT}(\bar{f}), \text{LT}(\bar{g})) \neq 1$;
- $\text{LT}(\bar{f}), \text{LT}(\bar{g}) > 1$.

In fact, if one of the previous condition holds, then the S -polynomial related to π reduces trivially to 0 under $\alpha \cap \beta$.

References

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- [2] A. Suzuki and Y. Sato. A simple algorithm to compute comprehensive Gröbner bases using Gröbner bases. In *Proceedings of the ISSAC 2006*, page 326–331, New York, NY, USA, 2006. Association for Computing Machinery.